

L1.3 HW/SW Platforms, Ethics Quiz

1. How is automotive E/E architecture evolving according to the presentation?

- A) From about 80-100 distributed ECUs toward a few high-performance ECUs plus many small sensor/actuator-interface ECUs
- B) From centralized computing toward hundreds of independent high-performance computers
- C) From electronic control toward entirely mechanical control
- D) From wired ECUs toward eliminating all in-vehicle networks

ANS:

2. What is one benefit of consolidating automotive hardware into a few centralized ECUs?

- A) It eliminates the need for sensors.
- B) It simplifies the architecture, reduces network load, and can improve reliability.
- C) It guarantees zero power consumption.
- D) It increases the number of independent networks required.

ANS:

3. Which technology supports high-performance automotive computing with a small form factor and low power consumption?

- A) Mechanical relays
- B) A trunk full of desktop computers
- C) System-on-Chip (SoC) technology
- D) An external data center as the only processor

ANS:

4. Why is hard real-time operation a design objective for autonomous vehicles?

- A) It reduces the number of vehicle doors.
- B) It guarantees that every algorithm uses a GPU.
- C) It makes software updates unnecessary.
- D) A missed deadline may compromise safety.

ANS:

5. What is identified as the most compute-intensive AV workload?

- A) Deep-learning inference
- B) Door-lock control
- C) Audio playback

D) Windshield-wiper control

ANS:

6. What is a typical automotive use of a Digital Signal Processor (DSP)?

- A) Global route planning
- B) Image preprocessing
- C) Mechanical braking
- D) Storing an HD map on paper

ANS:

7. Why are GPUs well suited to deep-learning workloads?

- A) They execute only one operation at a time.
- B) They contain no arithmetic units.
- C) They have many computational elements for highly parallel operations such as matrix multiplication.
- D) They are designed exclusively for body electronics.

ANS:

8. What is NVDLA in the NVIDIA DRIVE platform?

- A) An automotive Ethernet protocol
- B) A mechanical lidar scanner
- C) A general-purpose operating system
- D) An ASIC for deep-learning inference

ANS:

9. What is an FPGA?

- A) Reprogrammable hardware made of configurable logic blocks and programmable interconnections
- B) A fixed-function mechanical controller
- C) A satellite-based localization service
- D) A multimedia network protocol

ANS:

10. What is a drawback of using a large GPU batch for inference?

- A) It always lowers throughput.
- B) It can produce high and nondeterministic latency for an individual data item.
- C) It prevents SIMD computation.
- D) It converts the GPU into an FPGA.

ANS:

11. What is a key advantage of batch-less FPGA inference?

- A) It requires very large batches.
- B) It removes all hardware constraints.
- C) It can provide low and deterministic latency for any batch size.
- D) It is used only for neural-network training.

ANS:

12. Where can automotive FPGAs be deployed?

- A) Only in cloud data centers
- B) Only in infotainment displays
- C) Only in mechanical engine parts
- D) In smart sensors or as a central compute engine

ANS:

13. What are ASICs for deep learning often called?

- A) Neural Processing Units or AI accelerators
- B) Local Interconnect Networks
- C) Flash bootloaders
- D) Time-Sensitive Networks

ANS:

14. What is a major drawback of an ASIC compared with an FPGA?

- A) Lower specialization for its target computation
- B) Loss of programmability and flexibility
- C) A requirement to execute from a hard disk
- D) Inability to perform deep-learning inference

ANS:

15. Which platform was used for Tesla Autopilot HW1 from approximately 2014 to 2016?

- A) NVIDIA DRIVE PX2
- B) Tesla FSD HW4
- C) Mobileye EyeQ3
- D) Bosch AI Car Computer

ANS:

16. Which platform was used for Tesla Autopilot HW2?

- A) Mobileye EyeQ3

- B) Tesla HW4 FSD processor
- C) Google TPU
- D) NVIDIA DRIVE PX2

ANS:

17. Which sensor configuration is listed for Tesla HW4?

- A) Eight cameras
- B) Eight cameras, one radar, and twelve ultrasonic sensors
- C) Two cameras, one radar, and twelve ultrasonic sensors
- D) Six lidars and four radars

ANS:

18. How many CPU cores are listed for the Tesla HW3 FSD processor?

- A) 8
- B) 12
- C) 20
- D) 36

ANS:

19. Which NPU configuration is listed for Tesla HW4?

- A) One NPU at 10 TOPS
- B) Two NPUs totaling 36 TOPS
- C) Three NPUs totaling 50 TOPS
- D) No NPUs

ANS:

20. Which processors form the basis of the Delphi/Audi zFAS integration solution?

- A) Only a Google TPU
- B) Tesla HW3 and HW4 processors
- C) AMD Ryzen and Intel Xeon processors
- D) NVIDIA Tegra K1 and Mobileye EyeQ3

ANS:

21. Which in-vehicle network is positioned as the high-bandwidth backbone?

- A) Ethernet
- B) LIN

- C) Low-speed CAN
- D) MOST only

ANS:

22. Which network is intended primarily for multimedia audio/video transmission?

- A) FlexRay
- B) MOST
- C) LIN
- D) Low-speed CAN

ANS:

23. Which network is associated with safety-critical X-by-Wire applications?

- A) MOST
- B) LIN
- C) FlexRay
- D) Bluetooth

ANS:

24. Which network is used for body electronics such as doors, lights, and rearview mirrors?

- A) Ethernet TSN
- B) MOST150
- C) FlexRay
- D) LIN

ANS:

25. What data rate is listed for CAN-FD?

- A) 8 Mbps
- B) 10 kbps
- C) 150 Mbps
- D) 10 Gbps

ANS:

26. When was the AUTOSAR development partnership founded?

- A) 1995
- B) 2003

- C) 2015
- D) 2023

ANS:

27. For what type of ECU is AUTOSAR Classic Platform intended?

- A) Cloud-only servers
- B) Non-safety-critical smartphones
- C) Resource-constrained, safety-critical ECUs
- D) High-performance gaming computers

ANS:

28. For what type of ECU is AUTOSAR Adaptive Platform intended?

- A) Mechanical control systems with no software
- B) Only low-speed body-electronics nodes
- C) Resource-constrained 8-bit controllers only
- D) High-performance multicore automotive ECUs

ANS:

29. Which operating system standard is associated with AUTOSAR Classic?

- A) OSEK
- B) POSIX
- C) Android
- D) DDS

ANS:

30. Which communication style is associated with AUTOSAR Adaptive?

- A) Statically configured signal-based communication only
- B) Dynamically configured service-oriented communication using SOME/IP
- C) Mechanical point-to-point signaling
- D) ROS 1 Master-only communication

ANS:

31. What type of real-time constraint is listed for AUTOSAR Classic?

- A) No real-time constraint
- B) Best-effort only
- C) Hard real-time
- D) Soft real-time only

ANS:

32. Which platform should remain close to deterministic actuator and low-level vehicle-control functions?

- A) Android
- B) ROS 2 research tools
- C) AUTOSAR Adaptive
- D) AUTOSAR Classic

ANS:

33. What service does the ROS 1 Master node provide?

- A) Naming and registration services for other nodes
- B) GPU scheduling for CUDA kernels
- C) Flash-memory rewriting
- D) Vehicle-to-cloud routing

ANS:

34. What middleware does ROS 2 use for publish-subscribe communication?

- A) CAN-FD
- B) DDS
- C) MOST150
- D) OSEK

ANS:

35. Why is standard ROS difficult to certify to a high automotive safety level?

- A) It cannot communicate between nodes.
- B) It contains no software libraries.
- C) It relies on Linux as its underlying operating system.
- D) It requires AUTOSAR Classic for every message.

ANS:

36. To what Automotive Safety Integrity Level is the Apex.ai framework described as certified as a Safety Element out of Context?

- A) ASIL-A
- B) ASIL-B
- C) ASIL-C
- D) ASIL-D

ANS:

37. Which NVIDIA DRIVE component is the foundational stack containing an embedded RTOS, hypervisor, CUDA libraries, and TensorRT?

- A) DRIVE OS
- B) DRIVE IX
- C) DRIVE AV
- D) DriveWorks application only

ANS:

38. Which NVIDIA DRIVE component provides perception, mapping, and planning modules?

- A) DRIVE IX
- B) DRIVE AV
- C) MOST150
- D) AUTOSAR RTE

ANS:

39. On which ROS version is Autoware.AI based?

- A) DDS without ROS
- B) ROS 2
- C) ROS 1
- D) AUTOSAR Adaptive

ANS:

40. Which statement describes Baidu Apollo?

- A) A proprietary platform available only on NVIDIA hardware
- B) A Japanese platform based exclusively on ROS 1
- C) A body-electronics network protocol
- D) An open-source, hardware-neutral autonomous-driving platform from China

ANS:

41. What is the purpose of a flash bootloader in an ECU?

- A) To erase and rewrite flash memory for programming or software updates
- B) To accelerate matrix multiplication
- C) To transmit multimedia over MOST
- D) To perform object detection from cameras

ANS:

42. In an OTA update using a Connectivity ECU, when is the update loaded to the target ECU?

- A) Continuously while the vehicle is driving at full speed
- B) When the vehicle is in a safe state
- C) Only when all ECUs are powered off
- D) Before the software is wirelessly transferred

ANS:

43. Where is new software stored during an OTA update without a Connectivity ECU?

- A) Only in the driving application's active memory
- B) On a remote server with no local copy
- C) In a memory area separate from the driving application
- D) Inside an ultrasonic sensor

ANS:

44. In the classic trolley problem, what happens if the lever is pulled?

- A) The trolley stops with no casualties.
- B) The five people remain in danger and the side track stays unused.
- C) All six people are saved.
- D) The trolley moves to the side track, where one person is killed instead of five on the main track.

ANS:

45. Approximately how many decisions did the MIT Moral Machine platform gather?

- A) 40 million
- B) 400 thousand
- C) 4 billion
- D) 40 thousand

ANS:

46. How many factors were used to generate Moral Machine accident scenarios?

- A) Three
- B) Nine
- C) Twelve
- D) Twenty

ANS:

47. How many large cultural clusters are identified in the Moral Machine results presented?

- A) Two
- B) Four
- C) Three
- D) Nine

ANS:

48. Which preference was weaker in the Southern cultural cluster than in the other two clusters?

- A) Sparing lawful pedestrians over jaywalkers
- B) Sparing younger people over older people
- C) Sparing pedestrians over passengers
- D) Sparing humans over pets

ANS:

49. According to the presentation, what caused recent AV accidents rather than trolley-problem-style ethical choices?

- A) Failures in sensors or perception algorithms
- B) Deliberate choices to sacrifice passengers
- C) Cultural preferences programmed into the vehicle
- D) Moral Machine survey responses

ANS:

50. What does the presentation state about U.S. legislation for public-road AV testing?

- A) Every state has identical requirements.
- B) Regulation differs widely across states.
- C) Public-road testing is prohibited nationwide.
- D) Only federal law applies, with no state role.

ANS: